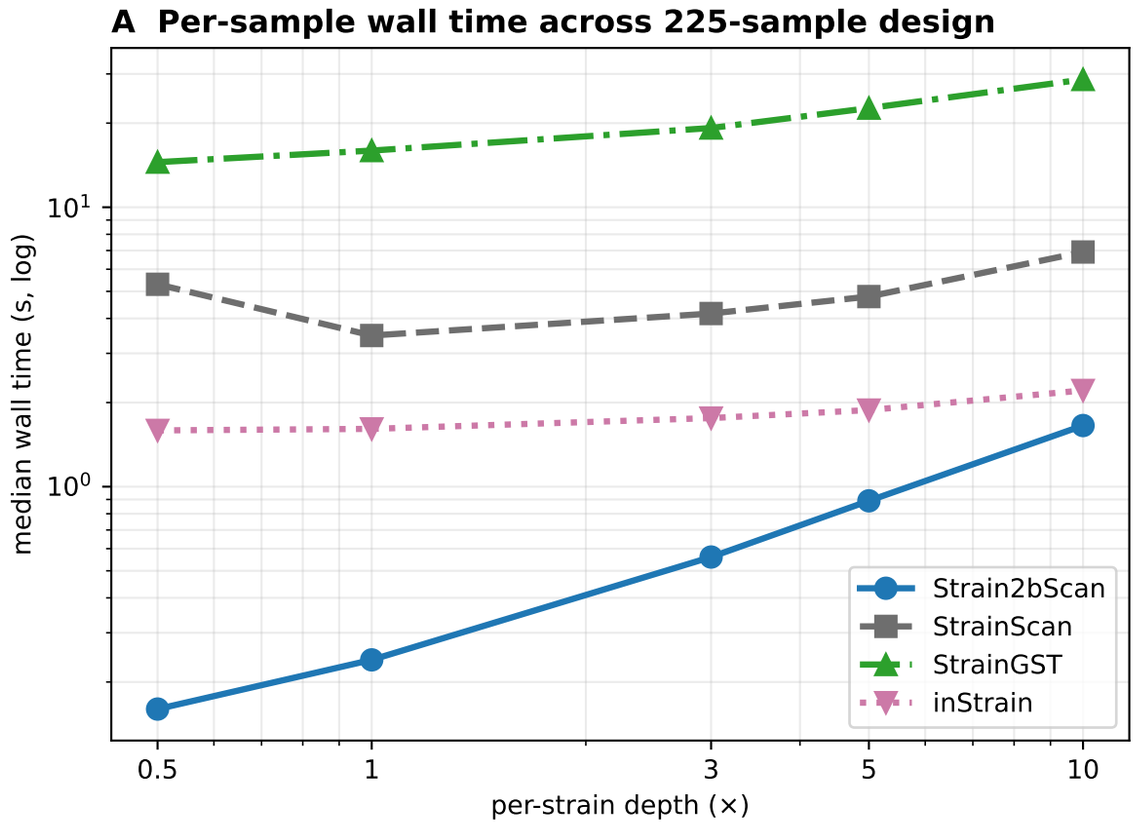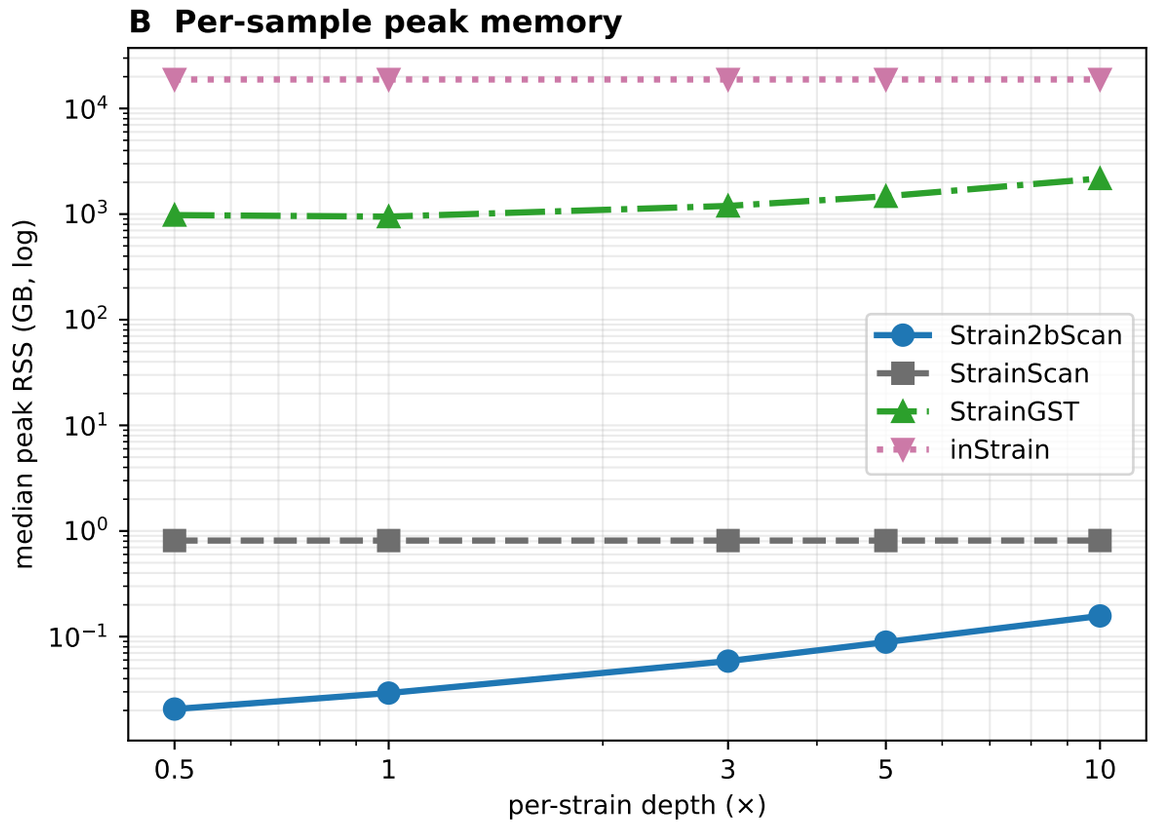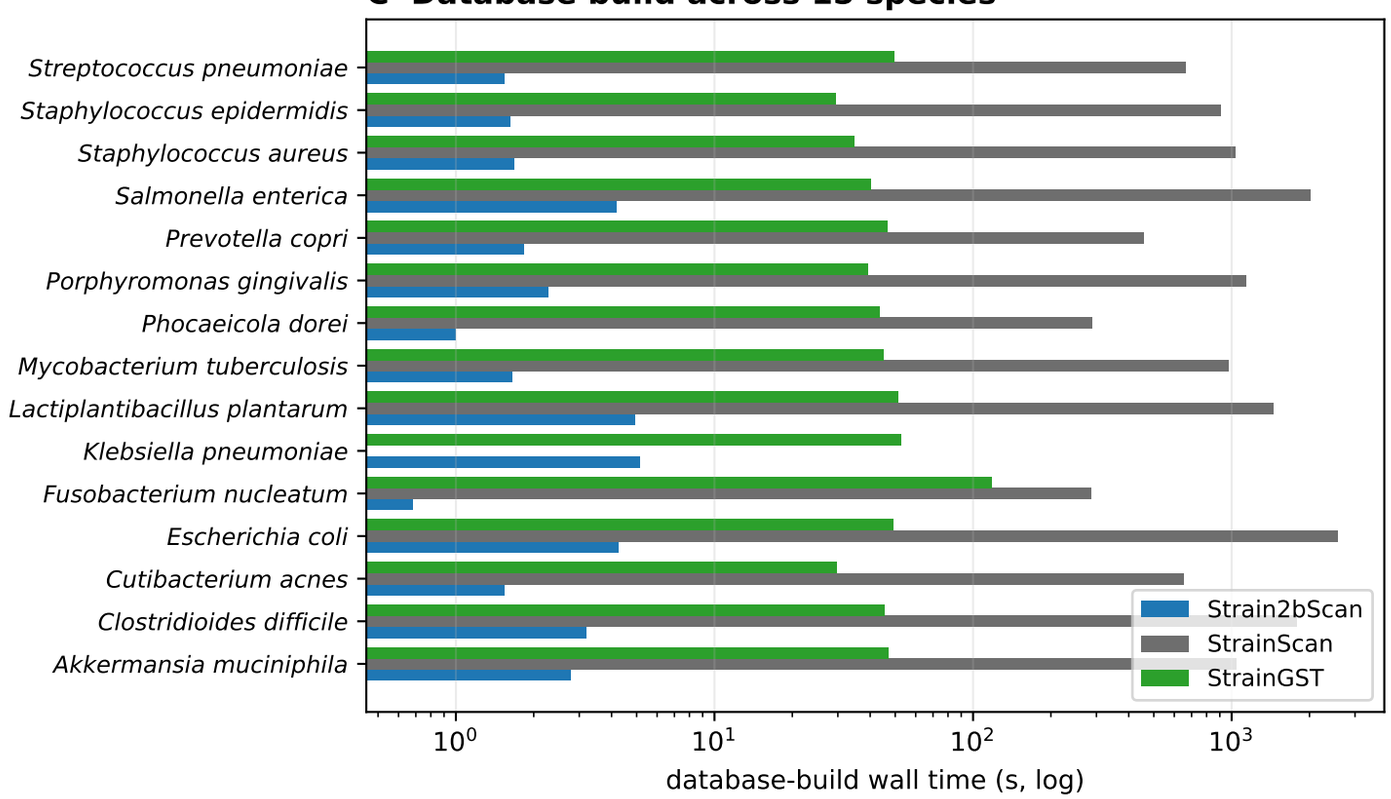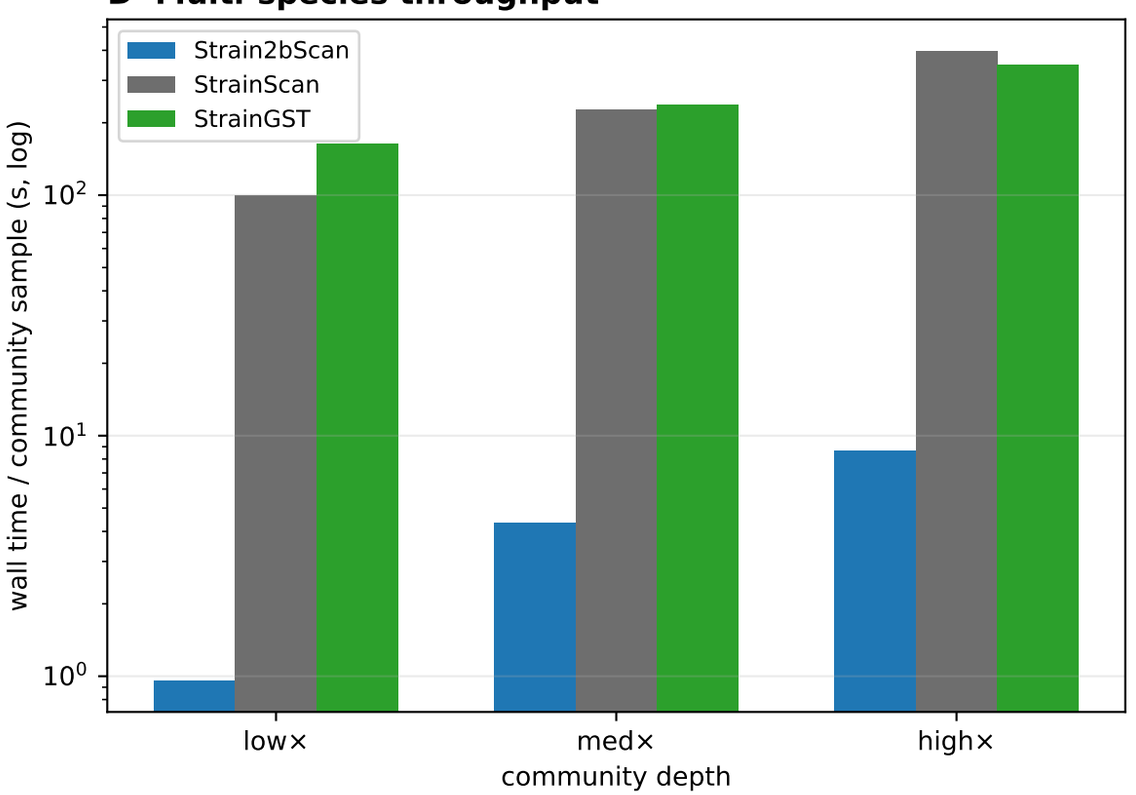Computational performance on the 15-species simulation benchmark